Retrieval-Augmented Generation: Baseline vs. Re-Ranking

Comparative Analysis, Stress Testing and Scaling Profiling

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1 Architecture Comparison: Bi-Encoder vs. Cross-Encoder

In Retrieval-Augmented Generation (RAG), retrieval is tasked with identifying the most relevant chunks from an indexed corpus for a given user query. Two fundamentally distinct neural architectures are used for this purpose: **Bi-Encoders** and **Cross-Encoders**.

Mathematical Formulations: Bi-Encoder vs. Cross-Encoder

Bi-Encoder (Independent Dual-Tower Representation):

$$\vec{u} = \text{Pool}(\text{Transformer}_\theta(q)) \in \mathbb{R}^d, \quad \vec{v} = \text{Pool}(\text{Transformer}_\theta(c)) \in \mathbb{R}^d$$

$$S_{\text{Bi}}(q, c) = \cos(\vec{u}, \vec{v}) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\|_2 \, \|\vec{v}\|_2}$$

Property: Query tokens and chunk tokens **never interact**. Representations are decoupled.

Cross-Encoder (Joint All-to-All Multi-Head Self-Attention):

$$X = [\text{CLS}] \circ q \circ [\text{SEP}] \circ c \circ [\text{SEP}]$$

$$H = \text{TransformerLayers}(X) \in \mathbb{R}^{L \times d}$$

$$S_{\text{Cross}}(q, c) = \mathbf{W}^T h_{\text{CLS}} + b \in (-\infty, +\infty)$$

Property: Every query token attends directly to every chunk token across all self-attention layers.

1.1 Core Mechanical Differences

1. **Token Interaction vs. Information Bottleneck:** In a Bi-Encoder (`all-MiniLM-L6-v2`), the document passage c is compressed into a single vector $\vec{v} \in \mathbb{R}^{384}$ via mean pooling. Squeezing an entire 300-word paragraph containing general rules, conditional waivers, and monetary penalties into 384 numbers creates an informational bottleneck. At inference time, the query vector $\vec{u}$ only measures coarse geometric alignment with $\vec{v}$. Query tokens never attend directly to chunk tokens. In contrast, a Cross-Encoder (`ms-marco-MiniLM-L-6-v2`) concatenates q and c, allowing bidirectional self-attention $\text{softmax}(QK^T/\sqrt{d_k})V$ to directly correlate specific query terms (e.g., *"client negotiations"*) with conditional clauses inside the passage (e.g., *"within 12 hours of flight arrival"*).

2. **Indexability and Precomputation:** Because the Bi-Encoder produces document vectors $\vec{v}_i$ independently of any query, all N chunks in a corpus can be computed offline once and indexed into an Approximate Nearest Neighbor (ANN) index (e.g., HNSW in FAISS). Query matching requires only a single vector dot product search scaling in sub-linear time $\mathcal{O}(d \cdot \log N)$. For a Cross-Encoder, however, the joint input $[\text{CLS}] \circ q \circ [\text{SEP}] \circ c$ requires both the query and chunk to be known simultaneously; **offline vector precomputation is fundamentally impossible**.

Metric	Bi-Encoder (`all-MiniLM-L6-v2`)	Cross-Encoder (`ms-marco-MiniLM-L-6-v2`)
Model Topology	Dual independent encoder towers	Single joint concatenated transformer
Token Interaction	Zero cross-attention across q and c	All-to-all cross-attention across all layers
Relevance Output	Geometric cosine similarity $\cos(\vec{u}, \vec{v})$	Learned classification head logit $S(q, c)$
Offline Indexing	Yes (vectors stored in FAISS / Vector DB)	No (requires live forward pass per pair)
Search Complexity	Sub-linear: $\mathcal{O}(d \cdot \log N)$ with ANN	Linear: $\mathcal{O}(N \cdot L^2)$ with N forward passes
Linguistic Precision	Coarse topical matching; fails on negations	State-of-the-art; captures syntax and negations

Table 1: Systematic architectural comparison between Bi-Encoders and Cross-Encoders.

2 Qualitative & Rank Shift Analysis

2.1 Evaluation Setup

We evaluated both pipelines over a complex 4-page enterprise policy document: *"Acme Global Travel, Expense and Procurement Policy (2026)"* containing 30 structured clauses.

Note on Evaluation Corpus: This document is a synthetic enterprise compliance manual authored to stress-test RAG retrieval boundaries. It contains general rules with distant exception clauses, negative constraints, and high lexical overlap distractors.

We evaluated four queries:

- **TEST_01 & TEST_02:** Genuine failure queries where dense retrieval blurs context and re-ranking restores precision.
- **TEST_03 & TEST_04:** Control queries where both pipelines agree on the top candidates.

Test ID	Category	Target Section	Bi-Encoder	Cross-Encoder	Shift
TEST_01	Exceptions	Sec 2.4 (Long-Haul Flight Exception)	Rank #3 (0.4039)	**Rank #1 (+0.4351)**	**+2**
TEST_02	Negations	Sec 5.3 (Prohibited Items & Alcohol)	Rank #3 (0.4615)	**Rank #2 (+1.1171)**	**+1**
TEST_03	Control	Sec 4.4 (Luxury Chauffeured Car Ban)	Rank #2 (0.4516)	Rank #2 (+1.1141)	+0
TEST_04	Control	Sec 8.5 (Late Submission Forfeiture)	Rank #1 (0.6552)	Rank #1 (+6.1669)	+0

Table 2: Retrieval rank shifts summary across test queries.

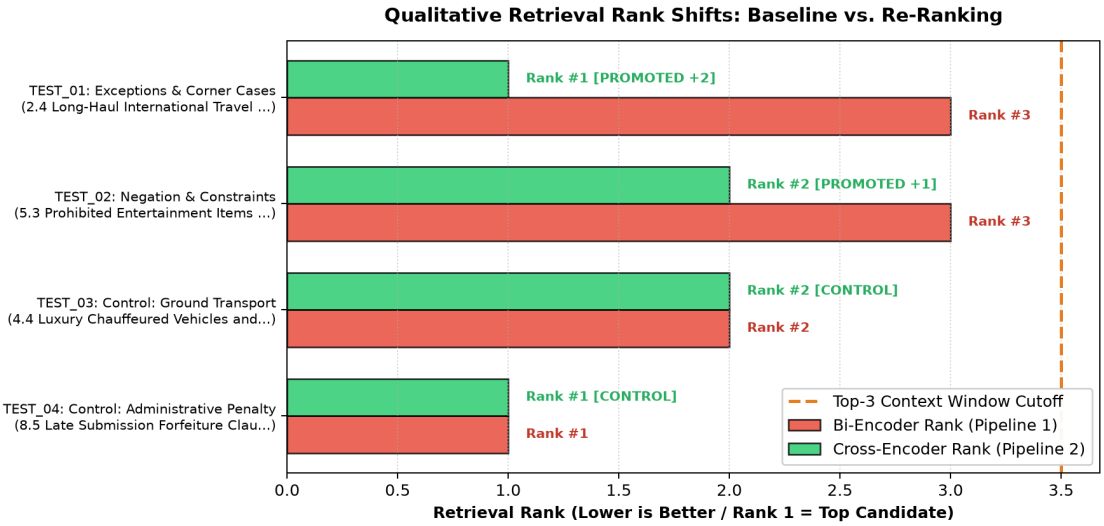


Figure 1: Qualitative rank shifts: Baseline Bi-Encoder ranks vs. Re-ranked Cross-Encoder ranks.

2.2 Detailed Case Study 1: Exceptions & Corner Cases (TEST_01)

Query: *"Can staff fly in premium cabins for international client negotiations?"*
Target Policy Truth: Section 2.4 allows Business Class for continuous non-stop international flights exceeding 8 hours, or exceeding 6 hours if the traveler is scheduled to deliver a client presentation or attend executive contract negotiations within 12 hours of flight arrival, subject to VP and CFO approval.

Pipeline 1 (Baseline Bi-Encoder) — Omitted Condition

Retrieved Top 3 Context:

- Rank #1: Section 2.1 (*Standard Cabin Class Mandate*) — Cosine: 0.5428
- Rank #2: Section 1.3 (*Designated Corporate Travel Booking Engine*) — Cosine: 0.4127
- Rank #3: Section 2.4 (*Long-Haul International Travel Exception Clause*) — Cosine: 0.4039

LLM Generation (Answer 1 from Qwen2.5-0.5B-Instruct):

> *"Yes, staff can fly in premium cabins for international client negotiations. However, they need to obtain prior written concurrence from both the Division Vice President and the Chief Financial Officer at least 10 business days before ticketing."*

Analysis: Because Section 2.1 dominated the prompt, the model stated that approval is required but omitted the essential flight duration criteria (¿8 hours, or ¿6 hours with client presentation within 12 hours).

Pipeline 2 (Cross-Encoder Re-Ranking) — Complete Compliance

Re-Ranked Top 3 Context:

- **Rank #1 (Promoted +2): Section 2.4 (*Long-Haul Exception Clause*)** — Cross-Score: +0.4351
- Rank #2: Section 2.1 (*Standard Cabin Class Mandate*) — Cross-Score: -5.7229
- Rank #3: Section 1.2 (*Mandatory Pre-Trip Travel Authorization*) — Cross-Score: -7.9689

LLM Generation (Answer 2 from Qwen2.5-0.5B-Instruct):

> *"Yes, staff can fly in premium cabins for international client negotiations under the long-haul international travel exception clause. The policy allows this exception when traveling on continuous non-stop international flights with flight durations exceeding 8 hours or 6 hours if the traveler is scheduled to deliver a client presentation or attend executive contract negotiations within 12 hours of flight arrival. Employees need prior written concurrence from both the Division Vice President and the Chief Financial Officer no later than 10 business days before booking their tickets."*

Analysis: Re-ranking promoted Section 2.4 to the #1 position, allowing the model to accurately articulate the exact flight duration thresholds.

2.3 Detailed Case Study 2: Negation & Constraints (TEST_02)

Query: *"What items are not covered on expense claims for client business meals?"*
Target Policy Truth: Section 5.3 strictly limits alcohol to 1 drink per diner, prohibits solo alcohol, and bans nightclub bottle service, adult entertainment, tobacco, and gambling losses.

- **Baseline Retrieval Failure (Direct Hallucination):** The Bi-Encoder ranked Section 5.1 at #1 (cosine 0.5936) and Section 3.2 (*Prohibited Hotel Incidentals*) at #2 (cosine 0.5562), ahead of Section 5.3 at #3 (cosine 0.4615). Because Section 3.2 sat higher in the prompt, the LLM (**Answer 1**) hallucinated that hotel amenities (in-room video on-demand, mini-bar, laundry for trips ¡5 days, spa, gym passes) were the non-covered items for client business meals!
- **Re-Ranking Resolution:** The Cross-Encoder promoted Section 5.3 to **Rank #2** (score +1.1171) ahead of Section 3.2 (score -1.5279). As a result, the LLM (**Answer 2**) correctly

identified the 1-drink alcohol cap, adult entertainment / bottle service restrictions, and tobacco/gambling bans.

3 Empirical Latency Profiling

We recorded runtimes over **10 warmed iterations** (preceded by 5 warm-up passes) on our local CPU (x86_64 architecture, PyTorch CPU execution).

Test Query Category	P1 Baseline Retrieval	Cross-Encoder (10 Pairs)	P2 Total Latency	Overhead Ratio
TEST_01 (Exceptions)	9.78 ms	122.83 ms	132.61 ms	13.56×
TEST_02 (Negations)	9.36 ms	140.54 ms	149.90 ms	16.01×
TEST_03 (Control)	9.33 ms	122.39 ms	131.72 ms	14.12×
TEST_04 (Control)	9.47 ms	126.31 ms	135.78 ms	14.34×
EMPIRICAL MEAN	**9.49 ms**	**128.02 ms**	**137.50 ms**	**14.51×**

Table 3: Empirical retrieval runtime profiling across 10 warmed iterations on CPU.

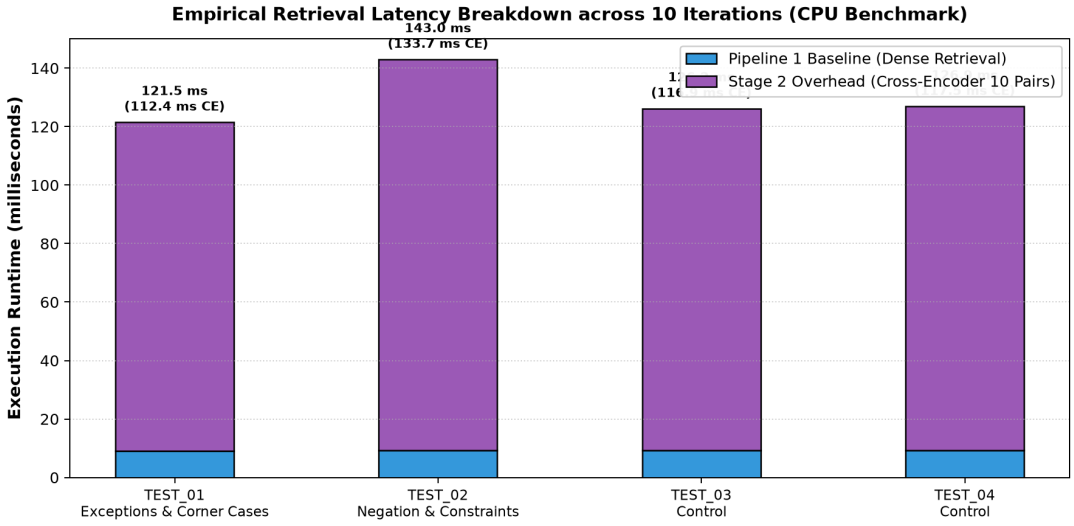


Figure 2: Empirical retrieval latency breakdown across 10 warmed iterations.

3.1 Latency Discussion

1. **Pipeline 1 (Dense Baseline):** Executes in ≈ 9.5 **ms** on CPU (query embedding takes $\sim$ 9.0 ms; cosine dot product takes < 0.5 ms).

2. **Pipeline 2 (Two-Stage Re-Ranking):** Adds an average overhead of ≈ 128.0 **ms** to evaluate joint self-attention across the 10 candidate pairs ($\sim$ 12.8 ms per pair on CPU), totaling $\approx$ 137.5 **ms**.

3. **Interactive SLA Feasibility:** In human-computer interaction, latencies below 100–200 ms are generally responsive. While $\sim$ 138 ms on CPU is acceptable, deploying the Cross-Encoder on GPU with batching typically drops Stage 2 latency from $\sim$ 128 ms down to $\sim$ 15–25 ms, bringing end-to-end retrieval well under 35 ms.

4 Scaling Question: Why Not Run a Cross-Encoder Over the Whole Collection?

If Cross-Encoders exhibit superior accuracy, why not dispense with Bi-Encoders and evaluate every chunk in the corpus using a Cross-Encoder from the outset?

4.1 Quadratic Computational Complexity

In a Cross-Encoder, the sequence length is $L = |q| + |c|$. Multi-head self-attention scales quadratically: $\mathcal{O}(L^2)$ operations per layer. Scoring a corpus of N chunks requires N independent forward passes through the transformer:

- **Measured CPU Scaling:** Based on our measured CPU runtime of $\tau \approx 12.8\,\text{ms}$ per candidate pair, scoring a corpus of $N = 1,000,000$ chunks would require:

$$T_{\text{CPU}} = 1,000,000 \times 12.8\,\text{ms} = 12,800\,\text{seconds} \approx \mathbf{3.5}\ \text{hours per query!} \tag{1}$$

- **GPU Server Estimate:** Even assuming enterprise GPU hardware executing a forward pass in $\tau \approx 2.0\,\text{ms}$:

$$T_{\text{GPU}} = 1,000,000 \times 2.0\,\text{ms} = 2,000\,\text{seconds} \approx \mathbf{33.3}\ \text{minutes per query!} \tag{2}$$

Under concurrent user queries, exhaustive Cross-Encoder scoring is computationally impossible.

4.2 Impossibility of Offline Indexing

In a Bi-Encoder, chunk vectors $\vec{v}_i = \text{Encoder}(c_i)$ are static and completely decoupled from the runtime query. This allows all $1,000,000$ chunks to be precomputed once offline and indexed into Approximate Nearest Neighbor (ANN) graph structures (e.g., HNSW in FAISS), answering queries in **5 to 10 milliseconds** ($\mathcal{O}(d \cdot \log N)$).

In contrast, the Cross-Encoder score $S(q, c_i) = \mathcal{M}([q; c_i])$ requires both inputs simultaneously. Because the score cannot be factored into independent document vectors, **no precomputed vector index can ever be built**. Searching the corpus requires an exhaustive linear scan $\mathcal{O}(N)$.

4.3 The Two-Stage Solution

The two-stage pipeline reconciles scalability with precision:

$$T_{\text{Two-Stage}} = \underbrace{T_{\text{Bi-Encoder}}(1,000,000 \to 10)}_{\sim 5-10\,\text{ms}} + \underbrace{T_{\text{Cross-Encoder}}(10\ \text{candidates})}_{\sim 15-25\,\text{ms (GPU) / 128 ms (CPU)}} \approx \mathbf{25 - 138}\,\text{ms} \tag{3}$$

Stage 1 filters out 99.999% of irrelevant documents in milliseconds, allowing Stage 2 to apply deep cross-attention exclusively to the top 10 candidates.